Digital Reader Chess Board

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by

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Abstract

Modern chess tournaments, as well as casual players, often demand a way to represent physical chess games digitally for streaming, analysis, game archiving, or other purposes. Solutions already exist; however, all of them employ complicated construction techniques and proprietary technology, making them very expensive and difficult to repair.

The *Digital Reader Chess Board* project converts physical chess piece positions using nothing more than a simple microcontroller and a tiny camera, leveraging intelligent computer vision techniques. Specifically, the camera in the *DRCB* searches for and identifies different colour combinations in predefined regions representing individual squares on the board. After a colour combination is identified, the microcontroller maps it to a specific chess piece ID and displays it on a screen.

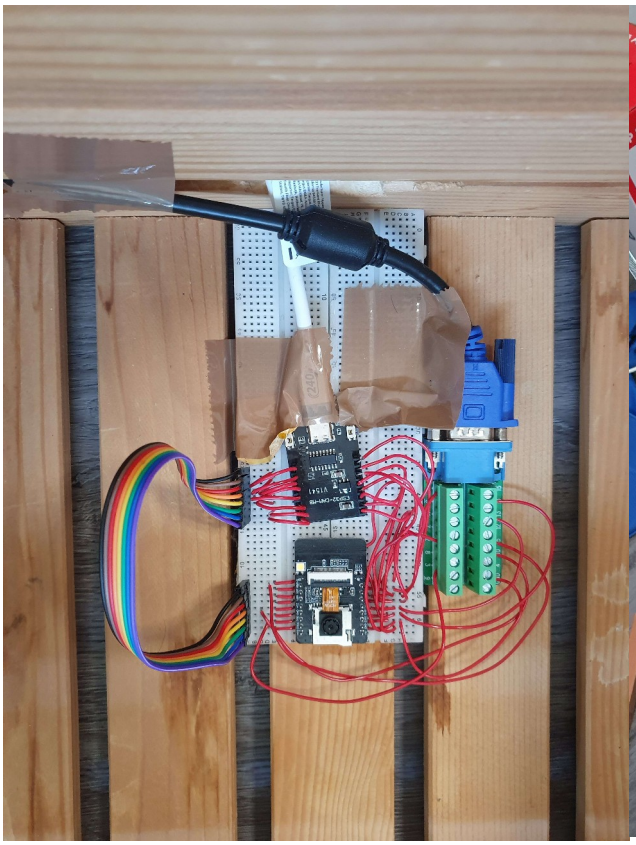
The result is a one-to-one, real-time digital mapping of an ongoing chess game. This digital reconstruction can naturally be used for game analysis, archiving, streaming, and other applications in the digital realm .However the reliability of the result is not up to tournament standard.

Introduction

Both professional chess tournaments and casual players increasingly require reliable ways to convert physical chess games into digital representations. A digital game state enables a wide range of applications, including live broadcasting, automated analysis, long-term archiving, and interaction with chess engines and other algorithmic systems.

A number of commercial solutions addressing this problem already exist. However, they tend to rely on highly specialised hardware, proprietary communication protocols, and complex mechanical or electronic designs. These characteristics lead to high costs, limited transparency, and effectively no user-side repairability or extensibility. As a result, such systems remain inaccessible to most users and poorly suited for experimental, educational, or open-ended development contexts.

The *DRCB* prioritises simplicity, affordability, and architectural transparency. The system is built using basic, low-cost materials, a rudimentary microcontroller, and a minimal camera module. Despite the severe hardware constraints, orders of magnitude less processing power and memory than modern consumer electronic (~1000 times less processing power and ~10000 times less memory than your typical smart fridge) the *DRCB* aims to achieve reliable, real-time translation of physical chess positions into a digital format.





Beyond its core functionality, the *DRCB* is explicitly designed as an open and extensible platform. Rather than locking users into predefined features, the system exposes its internal representation in a form that can be reused, modified, or expanded upon. This makes the project particularly well suited for educational use, experimentation with embedded computer vision, and future extensions such as alternative output formats, enhanced analysis, or integration with external systems.

In this report, the design decisions, methodological trade-offs, and technical implementation of the Digital Reader Chess Board are presented in detail. Both the successes and failures are examined.

Methodology

The primary methodological question regarding the *Digital Reader Chess Board* was reliably determining the position and identity of each chess piece using minimal, low-cost hardware. Two main fundamentally different approaches were taken into consideration: (1) detecting only whether a piece is present on a square and inferring its identity algorithmically, (2) assigning a unique identifier to each chess piece and reading that identifier directly.

Piece Identification Strategy

The presence-only approach has many advantages regarding hardware complexity. Under this model, the system would track moves from the initial board configuration and infer the identity of each piece based on legal move sequences. However, once a pawn reaches the final rank, the system cannot infer whether it was promoted to a queen, rook, bishop, or knight without explicit piece identification. Because there is no way to resolve this ambiguity algorithmically, the presence-only approach was rejected.

As consequence, the *DRCB* used the unique piece ID approach, where each chess piece carries some distinguishable detectable by a computer.

Alternative Identification Technologies

Several theoretically possible technological approaches to piece identification were considered, but rejected due to conflicts with the project’s design goals of simplicity, low cost, and repairability.

RFID-based systems were considered but dismissed due to increased hardware complexity. Each piece would require an embedded tag, and the board would need an array of readers, leading to communication overhead, signal interference, and a significant increase in system complexity.

Bluetooth-based identification suffers from similar drawbacks, but to an even greater extent. Power requirements, pairing complexity, and protocol overhead make Bluetooth unsuitable for a system designed to be simple and robust.

Pressure-based solutions using force sensors or pressure plates were also evaluated. While conceptually straightforward, sufficiently accurate sensors are expensive and difficult to calibrate, particularly for differentiating between multiple piece types, resulting in high cost and reduced reliability.

Magnetic identification was considered due to its low cost and simplicity. However, while magnets can reliably indicate the presence of a piece, encoding twelve distinct chess piece types (six per colour) in a robust and unambiguous manner proved impractical.

Camera-Based Computer Vision Approach

Though the process of elimination, a camera-based computer vision approach was chosen. This method requires only a single camera mounted beneath the playing surface and avoids per-piece electronics entirely. The board surface is divided into predefined square regions, and the system analyses each region independently.

Two visual identification strategies were evaluated: (1) pattern-based encoding and (2) colour-based encoding. Pattern-based approaches, such as QR-like markers, offer strong robustness but require significant image processing capability, which exceeds the practical limits of the chosen microcontroller.

Colour-based identification, while sensitive to lighting conditions, offers far lower computational complexity. To mitigate lighting sensitivity, the *DRCB* employs a composite colour approach in which each piece is identified by a combination of multiple colours. This increases robustness while keeping the image processing pipeline simple.

Hardware Platform and Output Constraints

The system is implemented on an ESP32-CAM microcontroller, selected for its low cost, integrated camera interface, and sufficient processing capability for lightweight computer vision tasks. The original design intended to output the reconstructed chess position via a VGA display. However, due to shared hardware resources and memory constraints, it was not possible to operate the camera and VGA output simultaneously on a single ESP32.

As a result, the output mechanism was redesigned. Instead of a local display, the ESP32 hosts a lightweight web server that provides a digital representation of the chessboard, accessible via a standard web browser. This approach not only resolves the hardware conflict but also enables remote access without additional equipment

Implementation

The implementation of the *DRCB* consists of 3 components: the physical board construction, the embedded hardware and the software architecture.

Physical Construction

The *DRCB* is built around a wooden superstructure that houses the electronic components and provides mechanical stability. Four metal bolts are embedded into the superstructure and act as supports for a transparent acrylic sheet, which serves as the playing surface. This elevated design allows the camera to be mounted beneath the board with an unobstructed view of the chessboard.

The chessboard grid itself is formed by tightly affixing rope onto the acrylic surface, creating clearly delineated square boundaries. This approach allowed us room for experimentation then we needed to make adjustments.

Each chess piece is fitted with a square cardboard base, slightly smaller than an individual board square. These bases carry the visual identifiers used by the computer vision system.

Embedded Hardware Integration

The electronic subsystem of the *DRCB* is centred around the ESP32-CAM microcontroller, which integrates both processing capabilities and a camera interface in a compact form factor. The camera is mounted centrally beneath the acrylic surface and aligned to capture the entire chessboard area in a single frame.

Power is supplied externally, and no additional sensors or peripherals are required for operation. All image acquisition, processing, and network communication tasks are executed locally on the microcontroller, ensuring full system autonomy once powered on.

At runtime, the system periodically captures images of the board and processes them to determine the current game state. The image processing pipeline operates on predefined board coordinates and treats each square as an independent region of interest.

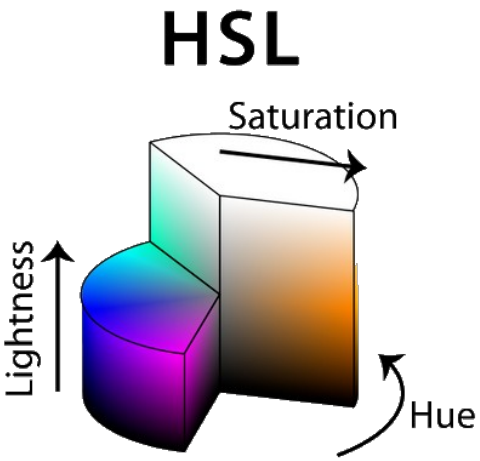
Digital Output and User Interface

Due to hardware resource constraints on the ESP32-CAM, simultaneous operation of the camera module and a VGA display was not feasible. As an alternative, the system exposes its internal board representation via a lightweight web server hosted directly on the microcontroller.

When accessed through a standard web browser, the server presents a graphical representation of the chessboard, updated in near real time. This approach eliminates the need for external display hardware while enabling remote access from multiple devices on the same network.

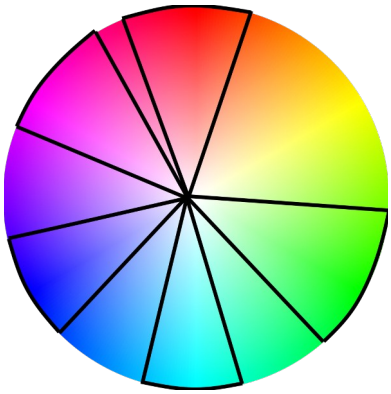
Computer Vision

The ESP32 allows us to capture images and store them in a buffer in RGB (red, green, blue) format. This is useful but RGB values are often unpredictably change in different lighting conditions. Our solution was to use a different format, namely HSL (hue, saturation, lightness). Saturation and lightness (both range from 0-100%) correspond to intensity of colour and its brightness respectably and hue (range from 0-360degrees) is the colour itself, saturation and lightness are effected by environment light but hue theoretically is independent from the environment light.



(HSL colour space, Wikipedia)

5 colours have enough combinations to represent all pieces. The figure below shows the colors we used along with their tolerances outlined in black.



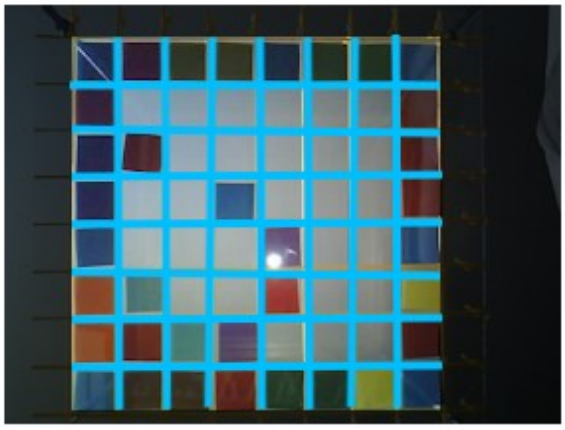
(Hues scanned for, authors own)

We wanted to avoid yellow with the assumption that usually tends to come from environment light, although by now it seems like this is a very minimal issue and its not unique to yellow.
With these colours we can define representations of all chess pieces:

	Black	White
Rook	red, dark blue	red, light blue
Bishop	red, pink	red, green
Knight	dark blue, light blue	dark blue, pink
Queen	dark blue, green	light blue, green
King	light blue, pink	pink, green
Pawn	pink	light blue

Thanks to the adjustable board, we can find a suitable physical configuration then hardcode the regions of the board. We then subdivide the board into 64 squares compare the hue of each pixel to our colors of interest and count how many match. If there is a significant amount of matches and they correspond to a piece, we assign that square with the piece.





To display to result we initially wanted to use a monitor with vga connection but due to a lack of pins, time and proper debugging we scrapped that idea and decided to host webserver on the ESP.

Evaluation

The D*igital Reader Chess Board* largely achieves its primary goal of translating a physical chess game into a digital representation. In several aspects, the *DRCB* performs better than initially expected. For instance, the materials required for the physical construction turned out to be significantly simpler and cheaper than anticipated, without negatively affecting stability or functionality. Another major success of the project is its full autonomy: once the system is set up, operating the chessboard merely requires powering it on, with no need for additional hardware or user intervention during normal operation.

As a prototype, however, the *DRCB* also exhibits a number of expected and unexpected drawbacks. One of the most significant limitations is the restricted processing power of the ESP32-CAM microcontroller. This constraint ultimately led to the abandonment of the original plan to display the reconstructed chessboard via a VGA output. Furthermore, the limited computational resources also restrict the number of clients that can simultaneously connect to the embedded web server in a stable manner.

The most problematic shortcoming of the current implementation lies in the colour-based piece identification system. While this approach was chosen for its simplicity and low computational cost, it proved to be highly sensitive to environmental conditions. Variations in lighting or surrounding colours can lead to incorrect piece identification, as external colours are sometimes misinterpreted as part of a piece’s identifier. In addition, the camera module and its associated software occasionally produce unstable readings, leading to sporadic misclassifications.

In summary, while the *Digital Reader Chess Board* successfully demonstrates the core concept of low-cost, camera-based chessboard digitisation, its current limitations highlight the trade-offs imposed by strict hardware constraints and environmental sensitivity. These issues clearly indicate areas for future improvement, particularly in sensing robustness and processing capability.